

A Venkata Vinesh Kumar Reddy

AI/ML Engineer | Software Developer | Reinforcement Learning & Quantitative Systems Enthusiast

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EDUCATION

Mahindra University, Hyderabad

B.Tech in Computer Science & Engineering | CGPA: 7.96 / 10.00

Expected: May 2027

Hyderabad, India

Strong Coursework: Design & Analysis of Algorithms, Probability & Statistics, Optimization Techniques, Machine Learning, Deep Learning, Reinforcement Learning, Financial Mathematics, Data Structures

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (ES6+), HTML5, CSS3, SQL

AI / Machine Learning: PyTorch, TensorFlow, Reinforcement Learning (PPO, DQN), Transformers, LSTMs, Weights & Biases (W&B;)

Software Engineering: Data Structures, Design & Analysis of Algorithms, OOP, Linux, Git Version Control

Mathematics & Analytics: Stochastic Calculus (SABR, Heston Volatility), Monte Carlo Simulations, Black-Litterman Optimization, Finite Difference pricing

Tools & Frameworks: FastAPI, React, Next.js, Tailwind CSS, Docker, WebSockets, Chart.js

TECHNICAL PROJECTS

Computational Finance & Exotic Derivatives Solver

C++, Python, NumPy, SciPy, OpenMP

- Problem:** Inefficiencies in pricing path-dependent exotic barrier derivatives under stochastic volatility using Python interpreters.
- Architecture & Dev:** Engineered a multi-threaded solver in C++ implementing the Crank-Nicolson Finite Difference Method and Monte Carlo path generators. Calibrated local-stochastic volatility parameters under Heston and SABR dynamics.
- Outcome:** Converged boundary value path equations in < 1.2ms, achieving an 8.4x pricing latency speedup over Python implementations with a 99.7% confidence interval.

Multivariate Time-Series & Deep Sequence Predictor

PyTorch, Python, Pandas, Weights & Biases

- Problem:** Capturing complex, non-linear atmospheric dependencies in multivariate weather forecasting datasets.
- Architecture & Dev:** Built a custom sequence modeling pipeline in PyTorch, configuring custom multi-head Attention Transformer encoders and stacked Recurrent LSTM/GRU layers. Leveraged Weights & Biases (W&B;) for loss optimization tracking.
- Outcome:** Reduced Mean Absolute Error (MAE) by 14.8% on global datasets compared to traditional statistical baselines (ARIMA/SARIMAX).

Veltrix Institutional AI Financial Terminal

React, JavaScript, Tailwind, Canvas, FastAPI

- Problem:** Sluggish telemetry updates in multi-widget financial UI dashboards rendering heavy charting and risk matrices.
- Architecture & Dev:** Designed a single-page React analytics console. Implemented custom HTML5 Canvas drawing engines to render real-time asset covariance grids, volatility smiles, and Value-at-Risk (VaR/CVaR) risk indices.
- Outcome:** Successfully modularized 12 interactive analytics grids in a single viewport, maintaining stable < 45ms UI redraw rendering latencies.

LEADERSHIP & EXTRACURRICULARS

Operations Lead

TEDx Mahindra University | 2024

- Spearheaded operational pipelines and logistics for a 500+ audience event. Coordinated cross-functional teams to manage timeline execution.

Logistics Head

Mahindra University Math Club | 2023

- 2024

- Engineered logistics, sponsorships, and scheduling for university-wide math modeling competitions and algorithmic puzzle contests, engaging over 300+ students.

International Chess Competitor

Bangkok Open | Teplice Open | 2024

- Competed at FIDE-rated international opens, displaying advanced strategic calculation, game theory, and rapid decision-making under intense competitive environments.